Brake Roller Tester



Category: **Automotive** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Brake roller tester

What it is used for

A brake roller tester checks vehicle braking performance by measuring the braking force produced at the wheels on rotating rollers. Correct vehicle positioning, controlled testing, and accurate recording help assess brake condition while reducing equipment and vehicle risks.

Main parts

- Roller assemblies rotate the vehicle wheels during testing.
- Drive motors power the roller assemblies.
- Force sensors or load cells measure braking force.
- Display or control console shows wheel force and test results.
- Vehicle positioning guides help align the wheels with the rollers.
- Safety guards cover moving parts where provided.
- Emergency stop controls halt the test system.
- Test software or recording system stores measurements on equipped models.

Before you start

1. Only trained and authorised personnel should operate the brake roller tester.
2. Read the tester and vehicle instructions and confirm the vehicle is compatible with the equipment.
3. Inspect rollers, guards, sensors, controls, and the test platform for damage or obstruction.
4. Check tyre condition, vehicle weight limits, and any special vehicle test requirements.
5. Ensure the test area is clear and the emergency stop is accessible.
6. Confirm the tester is ready, calibrated or verified as required, and free of fault indicators.

How to use

1. Guide the vehicle slowly onto the rollers following the tester's alignment instructions.
2. Position the test wheels correctly and apply the prescribed vehicle restraint or parking procedure.
3. Confirm that the vehicle transmission and driver controls are set as required by the test method.
4. Start the test using the console and allow the rollers to operate as instructed.
5. Apply the service brake gradually and consistently when prompted by the test procedure.
6. Observe and record braking force, left-right differences, and any displayed imbalance or fault values.
7. Repeat the test for the required axle or braking system according to the approved procedure.
8. Stop the rollers completely before moving the vehicle off the tester.

9. Drive the vehicle off slowly when the operator confirms it is safe to do so.
10. Save the results, report abnormal readings, and clear the test area after the vehicle exits.

⚠ Safety

- Keep people clear of rollers, moving wheels, and vehicle travel paths during testing.
- Never step onto rotating rollers or reach into moving mechanisms.
- Follow the tester's instructions for vehicle restraints, transmission position, and brake application.
- Do not test vehicles with unsafe tyres, leaking fluids, or other conditions prohibited by the procedure.
- Use the emergency stop if the vehicle shifts unexpectedly or a dangerous condition develops.
- Do not move the vehicle until the rollers have stopped and the operator authorises exit.

Care & storage

- Remove dirt and debris from the platform and accessible roller areas after use.
- Inspect roller surfaces, guards, sensors, and emergency stop controls regularly.
- Keep the test software and measurement system maintained according to the manufacturer instructions.
- Arrange authorised calibration and servicing if readings become inconsistent or faults occur.

Common mistakes to avoid

- Misaligning the vehicle wheels on the rollers.
- Applying the brake abruptly instead of following the test procedure.
- Ignoring vehicle compatibility, tyre condition, or displayed fault warnings.
- Driving off before the rollers stop and the operator authorises exit.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.